

Terceira Lista de Quantica:

① $L^2(\mathbb{R})$ o espaço de $\Psi: \mathbb{R} \rightarrow \mathbb{C}$ e $\hat{p} = -i\hbar \frac{d}{dx}$

$$\mathcal{D}(\hat{p}) = \{\Psi \in L^2(\mathbb{R}) \mid \Psi' \in L^2(\mathbb{R})\}$$

todo $\Psi \in L^2(\mathbb{R})$ e $\lim_{x \rightarrow \pm\infty} \Psi(x) = 0$. Mostre que $\hat{p}^\dagger = \hat{p}$

$$\text{Seja } \langle f | g \rangle = \int_{-\infty}^{\infty} f(x)^* g(x) dx$$

Queremos ver se $\langle \phi | \hat{p} \Psi \rangle = \langle \Psi | \hat{p}^\dagger \phi \rangle$

$$\int_{-\infty}^{\infty} \phi^* \left(-i\hbar \frac{d\Psi}{dx} \right) dx = \int_{-\infty}^{\infty} \left(-i\hbar \frac{d\Psi}{dx} \right)^* \phi dx = \int_{-\infty}^{\infty} i\hbar \frac{d\Psi}{dx} \phi dx$$

Usando $\int u dv = uv - \int v du$

$$\int i\hbar \frac{d\Psi}{dx} \phi dx = i\hbar \phi \Psi \Big|_{-\infty}^{+\infty} - i\hbar \int \frac{d\phi}{dx} \Psi dx$$

Note que por $\hat{p} = \hat{p}^\dagger$ $\phi \Psi \Big|_{-\infty}^{+\infty} = 0$ ou ainda $\phi \Big|_{-\infty}^{+\infty} = 0$.

② $\hat{T}(\alpha) = e^{-\frac{i\alpha}{\hbar} \hat{p}}$ mostre que $\langle x | \hat{T}(\alpha) | \Psi \rangle = \Psi(x-\alpha) = e^{-\alpha \frac{d}{dx}} \Psi(x)$

$$\hat{T}(\alpha) \approx \mathbb{I} - \frac{i\alpha}{\hbar} \hat{p}$$

$$\hat{T}(\alpha) | \Psi \rangle = \left(\mathbb{I} - \frac{i\alpha}{\hbar} \hat{p} \right) | \Psi \rangle$$

$$= \mathbb{I} | \Psi \rangle - \frac{i\alpha}{\hbar} \hat{p} | \Psi \rangle = \left(1 - \alpha \frac{d}{dx} \right) \Psi(x)$$

e atuando com $\langle x |$

$$\Psi(x-\alpha) = \langle x | \hat{T}(\alpha) | \Psi \rangle = \Psi(x) - i\alpha \frac{d\Psi}{dx} = \Psi(x) - \alpha \frac{d\Psi}{dx}$$

$$\Psi(x-\alpha) = e^{-\alpha \frac{d}{dx}} \Psi(x)$$

$$\textcircled{3} \hat{T}(a) \hat{X} \hat{T}^\dagger(a) = \hat{X} - a$$

atuando em um Ψ

$$\hat{T}^\dagger(a) = T(-a) \Rightarrow \hat{T}(a) \hat{X} (\hat{T}^\dagger(a) \Psi(x)) = \hat{T}(a) \hat{X} \Psi(x+a)$$

$$\text{e } \hat{X} \phi(x) = x \phi(x)$$

$$\begin{aligned} \hat{T}(a) (\hat{X} \Psi(x+a)) &= \hat{T}(a) (x+a) \Psi(x+a) \\ &= (x+a) \hat{T}(a) \Psi(x+a) \\ &= (x+a) \Psi(x) \end{aligned}$$

$$\begin{aligned} \text{Por outro lado } (x+a) \Psi(x) &= (\hat{X} + a) \Psi(x) = \hat{X} \Psi(x) + a \Psi(x) \\ &= x \Psi(x) + a \Psi(x) \\ &= (x+a) \Psi(x) \end{aligned}$$

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$$\textcircled{4} \begin{array}{ll} \{ |a_i\rangle \} & \text{base } \mathcal{H}_1 \quad A_{ij} \\ \{ |b_i\rangle \} & \text{base } \mathcal{H}_2 \quad B_{\alpha\beta} \end{array}$$

$\hat{A} \otimes \hat{B}$ que atua em $\mathcal{H}_1 \otimes \mathcal{H}_2$. obtenha os elementos de matriz desse operador na base $\{ |a_i, b_i\rangle \}$.

$$|a_i\rangle \otimes |b_i\rangle$$

Precisamos calcular:

$$\langle a_i, b_i | \hat{A} \otimes \hat{B} | a_j, b_j \rangle \quad \textcircled{1}$$

$$\text{e } \hat{A} \otimes \hat{B} (|u\rangle \otimes |v\rangle) = \hat{A} |u\rangle \otimes \hat{B} |v\rangle$$

$$\text{Logo de } \textcircled{1} : \langle a_i, b_i | \hat{A} | a_j \rangle \otimes \hat{B} | a_j \rangle$$

Temos então $\hat{A}|a_i\rangle = \sum_j A_{ji}|a_j\rangle$

Logo:

$$\langle a_i, b_i | \sum_{\xi, \gamma} A_{\xi \gamma} |a_\xi\rangle \otimes B_{\gamma i} |a_\gamma\rangle = \langle a_i, b_i | A_{\xi \gamma} B_{\gamma i} |a_\xi a_\gamma\rangle$$

Logo:

$$A_{\xi i} B_{\gamma i} \langle a_i, b_i | a_\xi, b_\gamma \rangle$$

O exercício nos dá $(\langle \phi_1 | \langle \phi_2 |) (| \phi_3 \rangle | \phi_4 \rangle) = \langle \phi_1 | \phi_3 \rangle \langle \phi_2 | \phi_4 \rangle$

$$\begin{aligned} p/ \quad \phi_1 &= a_i & \phi_3 &= a_\xi \\ \phi_2 &= b_i & \phi_4 &= b_\gamma \end{aligned}$$

$$= \langle a_i, a_\xi \rangle \langle b_i, b_\gamma \rangle$$

$$= \underline{\quad \quad \quad}$$

$$A_{\xi i} B_{\gamma i} \delta_{i\xi} \delta_{i\gamma} = A_{\xi \gamma} B_{ii}$$

ⓑ Como $(A^\dagger)_{ij} = A_{ji}^*$

$$\begin{aligned} \text{e } (\hat{A} \otimes \hat{B})^\dagger_{\kappa\alpha \xi j} &= (A_{\xi \gamma} B_{\gamma j})^* = (\hat{A} \times \hat{B})_{\xi j \kappa \alpha}^* = (A_{\xi \kappa} B_{ja})^* = A_{\xi \kappa}^* B_{ja}^* \\ &= A^\dagger \otimes B^\dagger \end{aligned}$$

ⓒ $(\hat{A} \times \hat{B})(\hat{C} \otimes \hat{D})$

$$(\hat{A} \otimes \hat{B}) \hat{C} \otimes \hat{D} |a\rangle \otimes |b\rangle = (\hat{A} \otimes \hat{B}) \hat{C} |a\rangle \otimes \hat{D} |b\rangle$$

$$= \hat{A} \hat{C} |a\rangle \otimes \hat{B} \hat{D} |b\rangle$$

$$= \underbrace{(\hat{A} \hat{C} \otimes \hat{B} \hat{D})}_{\neq} (|a\rangle \otimes |b\rangle)$$

$$\textcircled{d} \text{Tr}(\hat{O}) = \sum_n \langle n | \hat{O} | n \rangle$$

$$\begin{aligned} \text{Tr}(\hat{A} \otimes \hat{B}) &= \sum_{i,a} \langle a_i, b_a | \hat{A} \otimes \hat{B} | a_i, b_a \rangle = \sum_{i,a} A_{ii} B_{aa} = \\ &= \sum_i A_{ii} \sum_a B_{aa} = \\ &= \text{Tr}(\hat{A}) \text{Tr}(\hat{B}) \end{aligned}$$

$$\textcircled{e} \langle \vec{r} | \hat{p} \rangle$$

$$\hat{p} | p \rangle = p | p \rangle \Rightarrow \langle \vec{r} | \hat{p} | p \rangle = p \langle \vec{r} | p \rangle$$

$$\langle \vec{r} | \hat{p} | p \rangle = -i\hbar \nabla \langle \vec{r} | p \rangle = -i\hbar \psi_p(\vec{r})$$

Logo:

$$-i\hbar \nabla \psi_p(\vec{r}) = p \psi_p(\vec{r})$$

$$\psi_p(\vec{r}) = A e^{\frac{-ipr}{\hbar}} = \frac{1}{(2\pi)^{3/2}} e^{\frac{-ipr}{\hbar}}$$

$$1 = \int d^3p |p\rangle \langle p| = \int d^3p$$

Logo: de $\langle r | x \rangle = \psi(x)$

$$\langle \vec{r} | \int d^3p |p\rangle \langle p| | x \rangle = \psi(x)$$

$$\int d^3p \psi_p(\vec{r}) \psi_p(x) = \psi_x(\vec{r})$$

$$\int d^3p \frac{1}{(2\pi)^{3/2}} e^{\frac{-ipr}{\hbar}} \psi_p(x) = \psi_x(\vec{r})$$